ThermoRoute

Contents

ThermoRoute: a dynamic thermal-memory prior with calibrated, transferable multi-station river water-temperature forecasting

> Manuscript draft, rewritten around the large-sample evaluation. All numbers are produced by the code in this repository: the three-station case study by `scripts/04`, the 40-station large-sample experiment by `scripts/09` (5 seeds), and the calibration/decision/mechanism analysis by `scripts/10`. Reports: `outputs/reports/{usgs_experiment,usgs_analysis}.md`; figures `outputs/figures/`.

Abstract

Operational forecasts of daily river water temperature must respect two awkward facts: water temperature is so autocorrelated that simple persistence is a punishing baseline, and the apparent skill of many machine-learning studies depends on covariates unavailable at issue time. We develop ThermoRoute, which couples a learnable dynamic thermal-relaxation prior — a flow- and season-modulated generalisation of damped persistence toward climatology that contains the strong baseline as a special case — with a horizon-conditioned sparse variable–lag router and a bounded neural residual, and emits conformally-calibrated quantiles plus a high-temperature exceedance probability. We evaluate under a strict historical-information protocol on two settings. (i) A three-station regulated reservoir cascade (15 years), where we report the honest negative result that, because deep reservoir releases make water temperature near-perfectly persistent, no learned model improves on per-station damped persistence; the value there is confined to calibrated uncertainty and warnings. (ii) A 40-station large-sample set drawn from public USGS gages (free-flowing and regulated), where forecast headroom exists: ThermoRoute beats damped persistence at all three lead times (skill vs damped +0.13 / +0.06 / +0.03 at 1 / 3 / 7 days; better than damped at 81 / 92 / 86 % of stations) and beats persistence by +0.16 / +0.18 / +0.25, and is at least on par with a strong gradient-boosting baseline (per-station paired tests favour ThermoRoute at 3–7 days, $p<0.01$, and tie at 1 day). In 4-fold leave-group-out (every station held out once) it transfers to unseen basins, beating persistence by +0.13 / +0.14 / +0.23 (across-fold std ≈ 0.02) and damped persistence by +0.09 / +0.02 / +0.01. After conformal calibration its 90 % intervals are near-nominal (PICP ≈ 0.90). We deliberately report three negative results — no point-accuracy gain on the near-deterministic cascade, a flow-dependent thermal memory that does not generalise beyond it, and no robust cost–loss decision-value advantage over a (strong) deterministic persistence warning — and argue that the right scientific target is a calibrated, transferable forecaster whose advantage must be established on a large, hydrologically diverse sample rather than a single cascade.

Keywords: river water temperature; probabilistic forecasting; physics-guided machine learning; spatial transfer; conformal prediction; thermal inertia.

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1. Introduction

Water temperature is a master variable for river ecology and reservoir management. Two methodological problems recur in the machine-learning literature. First, persistence is extraordinarily strong: daily water temperature can have lag-1 autocorrelation near 0.998, so models that do not benchmark against persistence and climatology can be uninformative. Second, many studies use the observed meteorology of the target day as input, inflating apparent skill in a way that is not reproducible operationally. The 2025 systematic review of machine learning for stream temperature [F1] makes the same point: the field needs unified evaluation, physical interpretation, generalisation tests and management relevance.

We make three commitments and turn them into a testable design. (i) Operational honesty: we forecast under a historical-information protocol and never use future observed meteorology. (ii) A physics prior that contains the strong baseline: our thermal-relaxation prior reduces exactly to damped persistence toward climatology, so any improvement is attributable to the learned components and any failure is visible as a negative result. (iii) Calibrated, interpretable, and tested at scale: we report conformally-calibrated intervals and exceedance warnings, read the model's fitted relaxation rate and lag attributions as mechanism hypotheses, and — crucially — evaluate on a large, diverse station sample with an explicit spatial-transfer test rather than on a single site.

A central, honest finding shapes the paper: on a three-station regulated reservoir cascade, water temperature is so heavily damped that no learned model beats per-station damped persistence on point accuracy — the dynamic machinery helps only calibration and warnings there. We therefore move to a 40-station large-sample setting where forecast headroom exists, and show that ThermoRoute's value materialises: it beats the strong baseline on point accuracy, transfers across unseen basins, and produces near-nominally-calibrated warnings. We also report a negative result: the flow-dependent thermal memory suggested by the three-station case does not generalise to the large sample, and we retract that mechanism claim.

Contributions:

1. ThermoRoute, whose dynamic thermal-relaxation prior is a flow- and season-modulated generalisation of damped persistence, with a horizon-conditioned sparse variable–lag router and a bounded neural residual that cannot override the prior.
2. A leakage-audited evaluation with rolling-origin discipline, a one-shot blind test, moving-block-bootstrap confidence intervals, Diebold–Mariano tests, and an adversarial internal review of every headline claim.
3. A large-sample, transfer-tested demonstration: 40 public USGS stations, leave-group-out generalisation to unseen basins, and a unified point + probabilistic + event + decision-value assessment.
4. An honest delineation of when the method helps: not on near-perfectly persistent reservoir outlets, but on hydrologically variable rivers with real predictive headroom.

2. Data

Three-station cascade (case study). Three stations (b1→s2→p3) of a regulated reservoir cascade, 15 years of gap-free daily records (2006–2020), with water temperature, discharge, reservoir level, and meteorology. Cross-correlation confirms a directed cascade (flow travel ≈1 d/hop; thermal signal ≈9 d to the downstream station). Persistence is brutal here (lag-1 ≈ 0.998; full-record 2006–2020 persistence RMSE 0.24–0.36 °C at 1 day). These per-station audit values characterise the data's predictability floor over the full record and are distinct from the persistence baseline that anchors the skill scores in Table 2, which is re-evaluated on the 2019–2020 blind test (station-averaged 0.270 °C at 1 day, 0.918 °C at 7 days).

Large-sample USGS set (main). Forty stream gages retrieved programmatically from USGS NWIS (daily water temperature, discharge) with co-located Daymet meteorology (air temperature, precipitation, solar radiation as a physical radiative index, relative humidity), 2006–2020, selected for ≥55 % water-temperature coverage. The set spans free-flowing and regulated rivers across many states and a wide thermal range (≈ −1 to 31 °C). Crucially, it has real headroom: persistence 7-day RMSE has median ≈1.9 °C (range 0.9–3.3), versus full-record 0.79–1.23 °C at the reservoir outlets. Reservoir level and wind are unavailable at temperature gages, so the large-sample model uses a six-variable subset.

Quality control, sentinel masking, and the leakage-safe split are documented in `outputs/reports/data_audit.md` and `outputs/reports/usgs_acquisition.md`.

3. Methods

3.1 Problem and information set

For station s, issue day t, horizon h∈{1,3,7} predict `WTEMP_{s,t+h}` from information available at t only (Track H). No observation time-stamped after t enters the model.

3.2 Dynamic thermal-relaxation prior

Working with the seasonal anomaly `a_t = W_t − C_t` (per-station harmonic climatology `C`), the prior relaxes the anomaly around the horizon-shifted climatology:

```
e_t = g(weather_t) + b_s                                  (equilibrium anomaly)
κ   = σ(β_s + c_q·z(logFLOW) + c_l·z(WLEVEL) + w·season)  (daily relaxation rate)
â_h = e_t + (1−κ)^h (a_t − e_t)
Ŵ_{t+h}^prior = C_{t+h} + â_h
```

With `e_t=0` and κ=1−φ this is exactly damped persistence toward climatology; κ is warm-started near 0.05. Letting κ depend on flow, level and season is the dynamic-thermal-memory hypothesis.

3.3 Router, encoder, residual, heads

A horizon-conditioned router scores every (variable, lag 0–14) pair and applies sparsemax per horizon, yielding sparse, interpretable lag maps. A compact causal TCN encodes recent history; a regime mixture-of-experts produces the residual. The point forecast is `prior_h + Δ_h` where Δ is bounded by a **`tanh`** clamp so the prior is never overridden — added after an unbounded residual destabilised the hardest cascade station. The clamp is data-dependent: tight (±0.4 °C) on the small, strongly-damped cascade where the prior is the ceiling, and loose (±1.5 °C) on the large sample where headroom exists; the latter is selected on validation and is what lets the residual add skill at 3–7 days (§4.6). Monotone quantiles (`median ∓ softplus`) are trained with pinball loss; a separate head predicts the high-temperature exceedance probability. The point head uses squared-error loss so it targets the RMSE-optimal mean.

3.4 Conformal calibration

Per (station × horizon) split-conformal CQR on a held-out calibration year, with the exact ⌈(n+1)(1−α)⌉ order statistic and a calib/test boundary purge. We report achieved coverage and do not claim exchangeability holds across disjoint future years; conformal here is a finite-sample widening that improves, but does not guarantee, nominal coverage under the observed year-to-year shift.

3.5 Baselines and protocol

Persistence; damped persistence toward climatology; harmonic climatology; LightGBM (point + quantile + exceedance). Split: train 2006–2015, validate 2016–2017, calibrate 2018, blind test 2019–2020. All statistics fit on training data only. On the large sample, baselines and ThermoRoute are evaluated on identical windowed samples (observed targets only). Deep models use multiple seeds; we report the seed mean and, for the headline, the per-station win-rate and paired differences.

4. Results

4.1 Three-station cascade — an honest negative result

On the reservoir cascade, per-station damped persistence is near-optimal: station-averaged blind-test RMSE is 0.261 / 0.483 / 0.724 °C (1/3/7 d), and no learned model improves on it. ThermoRoute (joint, three seeds) is 0.343 / 0.557 / 0.808 °C — worse than damped persistence, significantly so at b1 and p3 and better only at s2 (Diebold–Mariano, Table 2b). LightGBM, GRU and the module ablations tell the same story; indeed the ablation that removes the mixture-of-experts matches damped persistence, indicating the extra machinery does not help point accuracy on this near-deterministic system. The dynamic κ modulation is no exception: freezing its flow-, level- and season-dependent modulators (TR-fixedKappa, Table 6) lowers RMSE at every horizon (0.287 / 0.532 / 0.788 vs 0.343 / 0.557 / 0.808 °C), so a constant per-station relaxation rate is at least as accurate as — and here slightly better than — the dynamic one; the dynamic-thermal-memory modulation thus earns its place on mechanism and interpretability grounds, not on point accuracy. The only value ThermoRoute adds here is probabilistic: conformal intervals (achieved PICP 0.65–0.91 across stations) and high-temperature warnings the point baselines cannot provide. We report this negative result in full rather than selecting a favourable framing, and use it to motivate the large-sample study: a single, heavily-damped cascade simply lacks the forecast headroom to distinguish models.

4.2 Large-sample USGS — ThermoRoute beats the strong baseline (Table A)

On 40 hydrologically diverse stations with real headroom, the picture inverts. Median over stations of per-station RMSE (identical samples; ThermoRoute = 5-seed mean):

Median over stations of per-station RMSE (5-seed mean; model uses 7 variables including gridMET wind, with the residual bound loosened to ±1.5 °C on the large sample — see §3.3):

horizon	persistence	damped	LightGBM	ThermoRoute	skill vs persist	skill vs damped	win-rate vs damped
1 d	0.671	0.645	0.560	0.554	+0.163	+0.127	81 %
3 d	1.420	1.258	1.153	1.175	+0.180	+0.057	92 %
7 d	1.952	1.525	1.458	1.490	+0.253	+0.034	86 %

Treating the 40 stations as the sample unit (the level at which we claim generality), per-station paired tests (Wilcoxon signed-rank; station-bootstrap 95 % CI on median skill; Table C) show ThermoRoute significantly beats persistence (median skill +0.16 / +0.18 / +0.25, $p \approx 10^{-10}$) and damped persistence (+0.13 / +0.06 / +0.03; 81 / 92 / 86 % of stations; $p < 10^{-6}$) at every horizon, with bootstrap CIs that exclude zero.

The comparison with a strong gradient-boosting baseline (LightGBM) is the interesting case, and we report it carefully because the aggregation matters. LightGBM has a marginally lower median RMSE at 3–7 days (1.153 / 1.458 vs 1.175 / 1.490) — driven by a handful of stations where it is much better. But at the station level ThermoRoute wins the head-to-head at 69 % of stations at both 3 and 7 days, and the per-station paired difference is significant (median skill +0.014 / +0.015, $p = 1\times10^{-3} / 7\times10^{-3}$); at 1 day the two are statistically tied (47 %, $p = 0.75$). So at a typical station ThermoRoute is at least as accurate as LightGBM and significantly better at the longer leads, while LightGBM retains a small edge in the station-averaged median. We therefore claim a robust improvement over the physics baselines and at-least-parity with a strong learned baseline, and we disclose both aggregations rather than choosing the flattering one.

4.3 Spatial transfer to unseen basins (Table B)

We use 4-fold leave-group-out so every one of the 40 stations is held out exactly once: each fold trains a station-agnostic model on the other 30 stations and forecasts the held-out 10. Averaged over folds, ThermoRoute beats persistence on the unseen basins by +0.13 / +0.14 / +0.23 skill at 1 / 3 / 7 days, with small across-fold variability (std ≈ 0.02; per-fold h7 skill 0.22–0.25), and it also beats damped persistence on the unseen basins (+0.09 / +0.02 / +0.01). This is the contribution a single-site study cannot make: the learned dynamic prior plus a station-agnostic residual generalises across basins, not just across years at one site.

4.4 Calibration, warnings and decision value

After conformal calibration, ThermoRoute's 90 % intervals achieve PICP 0.904 / 0.906 / 0.909 at 1 / 3 / 7 days on the large sample — essentially nominal, in contrast to the undercoverage on the three-station cascade (0.65–0.91). Coverage is also tight across stations: 97 / 92 / 89 % of the 40 stations fall within ±0.05 of the 0.90 target (Fig. `fig_usgs_calibration.png`), so the calibration is a population property, not a station-averaged artifact. The high-temperature exceedance warnings have modest positive skill (Brier-skill +0.30 / +0.25 / +0.24; AUPRC 0.57 / 0.51 / 0.49), comparable to LightGBM (+0.33 / +0.30 / +0.28).

We also report a negative result on decision value. On the cost–loss model the calibrated probabilistic warning does not robustly beat a deterministic persistence warning on this large sample: peak Relative Economic Value is 0.62 / 0.61 / 0.60 for ThermoRoute versus 0.89 / 0.80 / 0.73 for a persistence-threshold warning. The reason is hydrological — on these strongly-autocorrelated rivers "today already exceeds the threshold" is itself a strong predictor of future exceedance, so a deterministic persistence warning is a hard baseline. (An earlier, configuration-specific run suggested the opposite ordering; it did not replicate, and we report the robust finding.) The defensible probabilistic contribution is therefore the calibration (near-nominal coverage), not a decision-value advantage over persistence-based warnings.

4.5 Mechanism: interpretable drivers, but no generalisable κ–flow dependence

We report an honest negative result on the dynamic-memory hypothesis. On the three damped reservoir outlets the fitted relaxation rate κ rose ~2× from low to high flow (implied memory $1/\kappa \approx 10$–33 d), which we initially read as a flow-dependent thermal memory. This does not replicate on the large sample: across the 40 USGS stations κ rises with flow at only 24 % of stations (median $\kappa_{high}/\kappa_{low} = 0.93$; mean κ_{low} 0.113 vs κ_{high} 0.106), i.e. there is no consistent, physically directional flow dependence. Together with the ablation showing that freezing κ's modulators (TR-fixedKappa) does not worsen — and on the cascade slightly improves — point accuracy, we conclude that the dynamic-κ modulation is not a validated mechanism and not an accuracy lever; the three-station signal was a small-sample artifact. We therefore do not claim a flow-dependent thermal memory.

What does survive is interpretability of the router: its dominant variable shares shift sensibly with horizon — humidity- and air-temperature-led at 1 day (RHMEAN 35 %, TEMP 26 %, PRCP 15 %), with precipitation rising to share the lead at 7 days (RHMEAN 23 %, PRCP 22 %, TEMP 20 %) — consistent with event-driven, runoff-mediated temperature change becoming more important at longer leads. We present this as an interpretive read-out, not a causal mechanism.

4.6 Module ablations on the large sample

Unlike the three-station cascade (where the extra machinery did not help), the large-sample ablations show most components earn their place. We train each ablation at 3 seeds and compare it to the full model by a per-station paired test (median per-station RMSE at 1/3/7 d; Wilcoxon at h=3): removing the physics prior is catastrophic (0.995 / 1.302 / 1.531 vs the full 0.554 / 1.175 / 1.490; $p \approx 3\times10^{-11}$), confirming the prior carries the forecast — most dramatically at 1 day where RMSE nearly doubles; removing the mixture-of-experts hurts (0.623 / 1.236 / 1.527; $p \approx 6\times10^{-10}$); removing the router hurts slightly but significantly (0.568 / 1.189 / 1.497; $p \approx 2\times10^{-9}$). The exception is the dynamic-κ modulation: freezing it (TR-fixedKappa, 0.559 / 1.177 / 1.482) is within ±0.008 °C of the full model and is better at 7 days; although the per-station paired test is nominally significant ($p \approx 3\times10^{-5}$), the effect size is negligible and direction-inconsistent. The honest reading is that the prior, experts and router contribute real accuracy, while the dynamic-κ modulation is interpretive overhead we do not claim as an accuracy lever (consistent with §4.5).

5. Discussion

The two settings deliver a single message: the value of a physics-guided learned forecaster depends on whether the system has forecast headroom. On near-perfectly persistent reservoir outlets, damped persistence is a ceiling and the honest result is parity-minus on point accuracy with a calibration-only contribution. On a diverse large sample, the same model beats the strong baseline, transfers to unseen basins, and delivers near-nominally-calibrated warnings. This argues against single-site "state-of-the-art" claims and for large-sample, transfer-tested evaluation as the standard for this problem.

Limitations. The large-sample model omits reservoir level and wind (unavailable at temperature gages) and uses solar radiation as the radiative channel. The median margin over damped persistence at 3–7 days is small (the win-rate carries the result), and a strong gradient-boosting baseline (LightGBM) is competitive — slightly better than ThermoRoute at 3–7 days — so we claim a robust improvement over the physics baselines, not a new state of the art over all learners. We report five seeds. The dynamic-κ thermal-memory modulation does not generalise: the flow-dependence seen on three stations vanishes on the large sample, and freezing κ's modulators does not worsen RMSE, so we retract the mechanism claim and keep only the router's interpretable, horizon-dependent driver shares. We forecast under historical information; an operational-forcing track with archived weather forecasts would sharpen multi-day skill.

6. Conclusions

Under a strict historical-information protocol, ThermoRoute matches per-station damped persistence where the system is near-deterministic (a reservoir cascade) and beats the physics baselines where forecast headroom exists (a 40-station large sample), transferring to unseen basins and providing near-nominally-calibrated warnings. We deliberately report two negative results — no point-accuracy gain on the cascade, and no generalisable flow-dependent thermal memory — to keep the claims honest. The contribution is a calibrated, transferable, interpretable forecaster whose advantage over the physics baselines is established on a large, diverse sample rather than a single site, with the explicit caveat that a strong gradient-boosting learner remains competitive at the longer leads.

Data availability

This study uses two datasets.

(1) Three-station reservoir cascade (case study). Daily records (2006-01-01 to 2020-12-31; 5 479 days per station) of water temperature, discharge, reservoir level, air temperature, precipitation, wind speed, mean relative humidity and a radiative index (`DH`) at three cascade stations (b1, s2, p3). These were provided for this study; the `DH` field's data-dictionary definition is unconfirmed and we make no `DH`-specific physical claim. Two sentinel missing-codes (`WDSP=999.9`, `PRCP=99.99`; 0.1–0.3 % of records) are masked to NaN and imputed within the training fold.

(2) Forty-station USGS large sample (main analysis). Assembled programmatically from three open sources, in the same schema, by `scripts/data_usgs/build_usgs_stations.py`:

study variable	source	access	notes
WTEMP, FLOW, WLEVEL	USGS NWIS daily values	dataRetrieval/dataretrieval-python (U.S. Geological Survey); public domain	parameter codes 00010 / 00060 / 00065. Gage height (WLEVEL) is unavailable at most temperature gages (≈0 % coverage), so the large-sample model omits it; consequently the stage–discharge (rating-curve) physics is inactive at scale.
TEMP, PRCP, DH, RHMEAN	Daymet v4 (1 km gridded)	ORNL DAAC single-pixel API; open (CC0-equivalent for U.S. government / DAAC terms)	TEMP = mean of tmax/tmin; DH = incident solar radiation (srad, a physical radiative index replacing the original ambiguous DH, on a different scale); RHMEAN derived from vapour pressure via the Tetens saturation relation.
WDSP	gridMET	Climatology Lab / Northwest Knowledge Network THREDDS NetCDF-Subset Service; open	daily mean wind speed at the station coordinate.

Forty stream gages across 17 U.S. states (≥55 % water-temperature coverage over 2006–2020) span free-flowing and dam-regulated rivers; water-temperature coverage ranges 0.56–1.00 (median ≈0.85). The two assembled panels (`data_usgs/panel_usgs.parquet`, six variables; `data_usgs/panel_usgs_wind.parquet`, seven variables) and station metadata (`data_usgs/stations_meta.csv`, with USGS site numbers and coordinates) are included so the experiments reproduce without re-downloading; the raw per-site downloads can be regenerated by re-running the acquisition script. All three sources are public domain or open-licensed.

Code availability and reproducibility

All code, the fixed train/validation/calibration/blind-test boundaries, unit tests (leakage, splits, metrics, conformal), and one-command reproduction (`scripts/run_all.sh`) are in this repository. Per-day predictions, model weights and logs are regenerable by the staged scripts (`scripts/01–13`). An adversarial internal review (multiple independent expert lenses) checked every headline claim against the result tables; all negative results and ablations are retained, and a finding-by-finding disposition is provided in `outputs/reports/review_response.md`.

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